

What If You Could Taste Colors or See Music? Meet Synesthesia!

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Close your eyes and imagine the number 7. What color is it? For most people, it's just a number. But for some, the number 7 might always look bright yellow! This is called **synesthesia** (say: sin-uhs-THEE-zhuh). It's a **brain trait** where one sense triggers another. Someone with **synesthesia** might taste flavors when they hear music, or see shapes when they smell certain scents. It's not a sickness it's just a different way of experiencing the world. About 4 out of every 100 people have some form of it. One common type is **grapheme-color synesthesia, where letters and numbers have their own colors.** Another is **chromesthesia, where sounds create colors or shapes.** The famous painter Vincent van Gogh might have had this: he once said he saw colors when he played piano keys. Modern musician Billie Eilish also says she thinks of colors, textures, and even days of the week when she makes music. Scientists think synesthesia runs in families and is linked to our genes. It's like having extra connections in the brain that blend the senses together. So next time you hear a song, ask yourself: what color is it? You might just have a superpower you never knew about!

Word Treasure

synesthesia -- A brain condition where senses mix, like seeing colors when you hear music.

brain trait -- A natural characteristic of the brain, not an illness.

grapheme-color synesthesia -- A type where letters or numbers always appear to have specific colors.

chromesthesia -- A type where sounds, like music, make a person see colors or shapes.

genes -- Instruction codes inside your body that you inherit from your parents.

textures -- The feel or surface quality of something, like smooth or rough.

Background you need

Synesthesia was first studied scientifically over 100 years ago, but many people didn't believe it was real until recent decades.

Famous artist Vincent van Gogh (a Dutch painter known for *The Starry Night*) may have had chromesthesia, which influenced his vivid colors.

Today, musician Billie Eilish (an American pop star) says she experiences many senses mixing, which helps her create unique music and visuals.

Break it down

WHO: People with synesthesia, like Vincent van Gogh and Billie Eilish.

WHAT: Synesthesia, a brain trait that blends senses (like seeing colors for numbers).

WHERE: Inside the brains of about 4% of all people.

WHY: To show how some brains experience the world in amazing, unusual ways.

Why it matters

Learning about synesthesia teaches you that everyone's brain works differently, and some differences can be superpowers that make art and music even more amazing.

Test what you remember

1. What is synesthesia?

- ☐ (A) A brain trait that mixes senses
- ☐ (B) A type of music
- ☐ (C) A food
- ☐ (D) A sickness

2. Which example is given for grapheme-color synesthesia?

- ☐ (A) Tasting music
- ☐ (B) Number 7 always looking bright yellow
- ☐ (C) Hearing colors
- ☐ (D) Smelling shapes

3. What is chromesthesia?

- ☐ (A) When sounds create colors
- ☐ (B) When letters have colors
- ☐ (C) When smells trigger sounds
- ☐ (D) When tastes trigger shapes

4. Which famous painter might have had synesthesia?

- ☐ (A) Pablo Picasso
- ☐ (B) Vincent van Gogh
- ☐ (C) Leonardo da Vinci
- ☐ (D) Monet

5. According to scientists, what is synesthesia linked to?

- ☐ (A) Learning a language
- ☐ (B) Genes and family
- ☐ (C) Eating too much candy
- ☐ (D) Playing video games

6. How many people have some form of synesthesia?

- ☐ (A) 1 in 1000
- ☐ (B) 4 out of 100
- ☐ (C) Half of all people
- ☐ (D) Only one person

Answer key (try the quiz first!): 1: A 2: B 3: A 4: B 5: B 6: B

Questions to think about

1. **Positive:** Synesthesia can boost creativity, allowing people to see hidden patterns and connections that others miss.
2. **Critical:** Some people might think it's a disorder, and synesthetes could feel misunderstood or doubted.
3. **Scientific:** Researchers study synesthesia to better understand how the brain's sensory areas communicate and overlap.

MY THOUGHTS (at least 20 words):